

- (A) $\sin \theta = \frac{\nabla \varphi_1 \cdot \nabla \varphi_2}{|\nabla \varphi_1| |\nabla \varphi_2|}$ (B) $\cos \theta = \frac{\nabla \varphi_1 \cdot \nabla \varphi_2}{|\nabla \varphi_1| |\nabla \varphi_2|}$ 1 1 2
- (C) $\tan \theta = \frac{\nabla \varphi_1 \times \nabla \varphi_2}{|\nabla \varphi_1| |\nabla \varphi_2|}$ (D) $\cot \theta = \frac{\nabla \varphi_1 \times \nabla \varphi_2}{|\nabla \varphi_1| |\nabla \varphi_2|}$
7. A vector $\vec{r}$ is said to be solenoidal, if 1 1 2
- (A) $\text{curl } \vec{r} = 0$ (B) $\text{div } \vec{r} = 0$
- (C) $\text{curl } \vec{r} \neq 0$ (D) $\text{div } \vec{r} \neq 0$
8. The work done when a force $\vec{F} = (x^2 - y^2 + x)\vec{i} - (2xy + y)\vec{j}$ displaces a particle 1 2 2
in the xy -plane from $(0, 0)$ to $(1, 1)$ along the curve $y = x$ is
- (A) $-\frac{1}{3}$ (B) $\frac{2}{3}$
- (C) $-\frac{2}{3}$ (D) $\frac{1}{3}$
9. If $\mathcal{L}[f(t)] = F(s)$, then $\mathcal{L}[f(at)] =$ 1 1 3
- (A) $F(s + a)$ (B) $F\left(\frac{s}{a}\right)$
- (C) $\frac{1}{a}F\left(\frac{s}{a}\right)$ (D) $F(s - a)$
10. $L[te^{-t}] =$ 1 2 3
- (A) $\frac{1}{s - 1}$ (B) $\frac{1}{(s - 1)^2}$
- (C) $\frac{1}{(s + 1)^2}$ (D) $\frac{1}{(s - 1)^3}$
11. $L^{-1}\left[\frac{1}{s^2 + 25}\right] =$ 1 1 3
- (A) $\sin 5t$ (B) $\frac{1}{5}\sin 5t$
- (C) $\cos 5t$ (D) $\frac{1}{5}\cos 5t$
12. If $L[f(t)] = F(s)$, then $L^{-1}[F(s - a)] =$

(A) $e^{-at} f(t)$	(B) $\frac{1}{a} f\left(\frac{t}{a}\right)$	1	2	3
(C) $\int_0^t f(t) dt$	(D) $e^{at} f(t)$			
13. An analytic function with constant modulus is		1	1	4
(A) zero	(B) analytic			
(C) harmonic	(D) constant			
14. A transformation that preserves angles between every pair of curves through a point only in magnitude, but not in direction is said to be _____ at that point.		1	1	4
(A) isogonal	(B) conformal			
(C) regular	(D) irregular			
15. The invariant points of the transformation $w = \frac{2z+6}{z+7}$ are		1	2	4
(A) 6, -1	(B) 3, 2			
(C) -3, 2	(D) -6, 1			
16. If $w = f(z) = u + iv$ is an analytic function, then		1	2	4
(A) $\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}\right) f(z) ^2 = 4 f'(z) ^2$	(B) $\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}\right) f(z) = 4 f'(z) $			
(C) $\left(\frac{\partial}{\partial x} + \frac{\partial}{\partial y}\right) f(z) ^2 = 4 f'(z) ^2$	(D) $\left(\frac{\partial}{\partial x} + \frac{\partial}{\partial y}\right)^2 f(z) ^2 = 2 f'(z) ^2$			
17. A zero of an analytic function $f(z)$ is a value of z for which		1	1	5
(A) $f(z) = 1$	(B) $f(z) \neq 1$			
(C) $f(z) \neq 0$	(D) $f(z) = 0$			
18. By Cauchy's residue theorem, the value of $\oint_C \frac{z}{z-3} dz$ where C is the circle $ z = 2$ is		1	2	5
(A) 1	(B) $4\pi i$			
(C) 0	(D) $2\pi i$			
19. For $f(z) = \frac{\sin z}{z}, z=0$ is _____		1	1	5
(A) Removable singularity	(B) Pole of order 0			
(C) Essential singularity	(D) Pole of order 1			
20. The singular points of the function $f(z) = \frac{z+2}{(z+1)(z-3)}$ are		1	2	5
(A) 0, 3	(B) -1, 3			
(C) -2, 1	(D) 1, 3			

PART - B (5 × 8 = 40 Marks)

Answer all Questions

21. (a) Evaluate $I = \int_0^4 \int_{\frac{x^2}{4}}^{2\sqrt{x}} dy dx$ by changing the order of integration.	8	3	1
(OR)			
(b) Find the volume of the sphere $x^2 + y^2 + z^2 = a^2$ by using triple integration.			
22. (a) Show that $\vec{F} = (2x + yz)\vec{i} + (4y + xz)\vec{j} - (6z - xy)\vec{k}$ is irrotational and hence find its scalar potential.	8	3	2
(OR)			
(b) Evaluate Green's theorem in the plane for $\oint_C (x^2 + 2xy)dx + (y^2 + x^3y)dy$ where C is a square with vertices $P(0,0)$, $Q(1,0)$, $R(1,1)$ and $S(0,1)$.			

23. (a) Find $L^{-1} \left(\frac{1}{s(s^2 + a^2)} \right)$ by using convolution technique. 8 3 3

(OR)

- (b) Using Laplace transform, solve the initial value problem $y' + 3y = e^{2t}$, $y(0) = 1$.

24. (a) Prove that the function $u = \frac{1}{2} \log(x^2 + y^2)$ is harmonic. Find $f(z)$ by using Milne-Thomson method. Also find its harmonic conjugate. 8 3 4

(OR)

- (b) Find the bilinear transformation which maps the points $-1, 0, 1$ in the z -plane onto the points $0, i, 3i$ in the w -plane respectively.

25. (a) Evaluate $\oint_C \frac{z+1}{z(z-1)} dz$ where C is the circle $|z| = 2$ by using Cauchy's integral formula. 8 3 5

(OR)

- (b) Expand $f(z) = \frac{1}{(z-1)(z-2)}$ as Laurent's series valid in the regions: (i) $1 < |z| < 2$ (ii) $|z| > 2$.

PART - C ($1 \times 15 = 15$ Marks)

Answer any 1 Questions

Marks BL CO

26. Verify Gauss divergence theorem for $\vec{F} = (x^2 - yz)\vec{i} + (y^2 - xz)\vec{j} + (z^2 - xy)\vec{k}$ taken over the rectangular parallelepiped bounded by $0 \leq x \leq a$, $0 \leq y \leq b$ and $0 \leq z \leq c$. 15 4 2
27. Evaluate $\int_0^{2\pi} \frac{d\theta}{5+3\cos\theta}$ by using Contour integration. 15 4 5

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